GOLD: Debugging Programs Live (Fri Aug 28)

Flipped Friday #1

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How this hour works

- **iClicker:** today is still practice; participation tracking starts Monday.
- **Ed challenges**: the test case scores are just for practice; we never include these in the class participation.
- Cass will walk through opening a challenge, Run, and Test before we start.

Outline

- Warm-up poll (iClicker)
- Exercise 1: Fix the Receipt (~10 min)
- Exercise 2: Average of Three (~8 min)
- Review
- Close + one last poll
- If we finish early: *Optional: First If* (last slide)

Each challenge arrives with test 1 already green. Your job is the rest.

Warm-up poll

Answer in iClicker.

Q1. What does `IO.println("Items: " + 2 + 3);` print?

- **A.** `Items: 5`
- **B.** `Items: 23`
- **C.** It refuses to compile
- **D.** `Items: 2 3`

Warm-up (iClicker)

Question *Submitted Aug 28th 2026 at 3:38:36 pm*

What does this print?

```
IO.println("Items: " + 2 + 3);
```

- ( ) Items: 5
- (•) Items: 23
- ( ) It refuses to compile
- ( ) Items: 2 3

Friday 1: Fix the Receipt

Fix the Receipt

Press **Run** and try `2` burgers, `3` fries, and `30` paid.

The Total line prints nonsense.

Step 1: fix it

Step 2: uncomment the Change line. It will *not* compile. Read the compiler's message then fix that line the same way.

Test 1 already passes.

Friday 1: Average of Three

Average of Three

Read three whole numbers and print their average as a decimal: for `1`, `2`, `4` Java prints `Average: 2.3333333333333335`. Replace the `TODO` line.

Test 1 already passes; make the other two green.

If test 3 says you printed `2.0`, you have just met a very common Java bug. We'll fix it together on screen in a few minutes but try it yourself first.

A Common Wrong Answer

This is what many of you wrote, at least for your first attempt:

```
void main() {
    int a = Integer.parseInt(IO.readln("First: "));
    int b = Integer.parseInt(IO.readln("Second: "));
    int c = Integer.parseInt(IO.readln("Third: "));
    double average = (a + b + c) / 3;     // the modal wrong answer
    IO.println("Average: " + average);
}
```

For 1, 2, 4 it prints `Average: 2.0`.

Test 1 passes. Test 2 passes. But Test 3 fails.

Why?

Polls (iClicker)

Q2. `double avg = 7 / 2;` leaves `avg` holding `3.0`, not `3.5`. Why?

- **A.** Assigning to a `double` variable rounds the integer quotient up to the nearest tenth, producing `3.0`.
- **B.** `7 / 2` is evaluated with two `int`s first, giving `3`, and only that result is then converted to `3.0`.
- **C.** A `double` reserves its fractional digits for later input, so a value assigned at creation keeps only its whole part.
- **D.** The compiler sees the `.0` in the type `double` and matches the quotient to it by discarding any fractional result.

Q3. Which line makes the average of 1, 2, 4 print `2.3333333333333335`?

- **A.** `double average = 0.0 + (a + b + c) / 3;`
- **B.** `double average = (a + b + c) / 3.0;`
- **C.** `double average = (a + b + c) / 3;`
- **D.** `double average = a + b + c / 3.0;`

Let's fix it live!

▶ Expand

Why 2.0? (iClicker)

Question 1 *Submitted Aug 28th 2026 at 3:45:01 pm*

`double avg = 7 / 2;` leaves `avg` holding `3.0`, not `3.5`. Why?

- Assigning to a `double` variable rounds the integer quotient up to the nearest tenth, producing `3.0`.
- **`7 / 2` is evaluated with two `int`s first, giving `3`, and only that result is then converted to `3.0`.**
- A `double` reserves its fractional digits for later input, so a value assigned at creation keeps only its whole part.
- The compiler sees the `.0` in the type `double` and matches the quotient to it by discarding any fractional result.

Question 2 *Submitted Aug 28th 2026 at 3:45:23 pm*

Which line makes the average of 1, 2, 4 print `2.3333333333333335`?

- `double average = 0.0 + (a + b + c) / 3;`
- **`double average = (a + b + c) / 3.0;`**
- `double average = (a + b + c) / 3;`
- `double average = a + b + c / 3.0;`

Review

What we've done today

- You've debugged two programs already!
- On Monday: types and Strings, where the `+` and `/` traps get the full story.
- Not finished? The challenges stay open all semester. Finishing them is the cheapest exam prep you will ever do.
- Want one more? *Optional: First If* is the last slide.

Last poll (iClicker)

Q4. What happens? `int n = IO.readln("How many? ");`

- **A.** Runs; `n` holds the number typed
- **B.** Runs, but crashes if you type letters
- **C.** Refuses to compile: `String cannot be converted to int`
- **D.** Runs; `n` holds 0

Last Poll (iClicker)

Question *Submitted Aug 28th 2026 at 3:46:19 pm*

What happens?

```
int n = IO.readln("How many? ");
```

- ( ) Runs; n holds the number typed
- ( ) Runs, but crashes if you type letters
- (•) Refuses to compile: String cannot be converted to int
- ( ) Runs; n holds 0

Friday 1 (optional): First If

Optional: First If

Read in a whole-dollar total and a member flag (`1` = member, `0` = not), both as `int`s.

Members pay 10% less, in **whole dollars**: `total - total / 10` (integer division is fine here). Print one line, `Pay: $N`. Don't display any decimal places (no cents).

One `if` is all it takes. Test 1 already passes.